

Experiment 5

Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily leading to multiple integrations per day.

Each integration is verified by an automated build (including test) to detect integration errors as

quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building

and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of

tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and

features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins

Creating a job:

Naming the job and setting it as freestyle:

Selecting build type as “Execute shell”:

Entering a simple command for the shell execution:

Applying and saving the project configuration:

Building the project:

Console output (after building):

Example 1.2: Taking parameters through files

Contents of script example1.cmd:

Executing script example1.cmd on the terminal:

Modifying the Jenkins project to execute the script while supplying required parameters:

Console output after building the modified project:

Example 2

Example 2.1: Running a Java program under Jenkins

Creating a simple Java program:

Compiling and running the program on the terminal:

Creating a new freestyle project:

Configure new project:

Console output after building:

Example 3

Example 3.1: Parameterise build

Creating a new freestyle project:

Enabling parameterisation and adding a String parameter:

Configuring the string parameter as Fname:

Adding a choice parameter and configuring it as City with the following choices:

Creating a script which takes 2 arguments for name and city:

Configuring build steps:

Entering parameters for build:

Console output after building:

Example 3.2: Running a Java program with parameters

Creating a Java program with an input argument:

Testing the program on the terminal:

Creating a new freestyle project:

Parameterise the project by adding a string parameter as follows:

Configure the build steps:

Entering the parameter for the build:

Console output after building:

Example 5

Example 5.1: Running a Python program

Creating a simple Python script:

Running the Python script on the terminal:

Creating a new freestyle project:

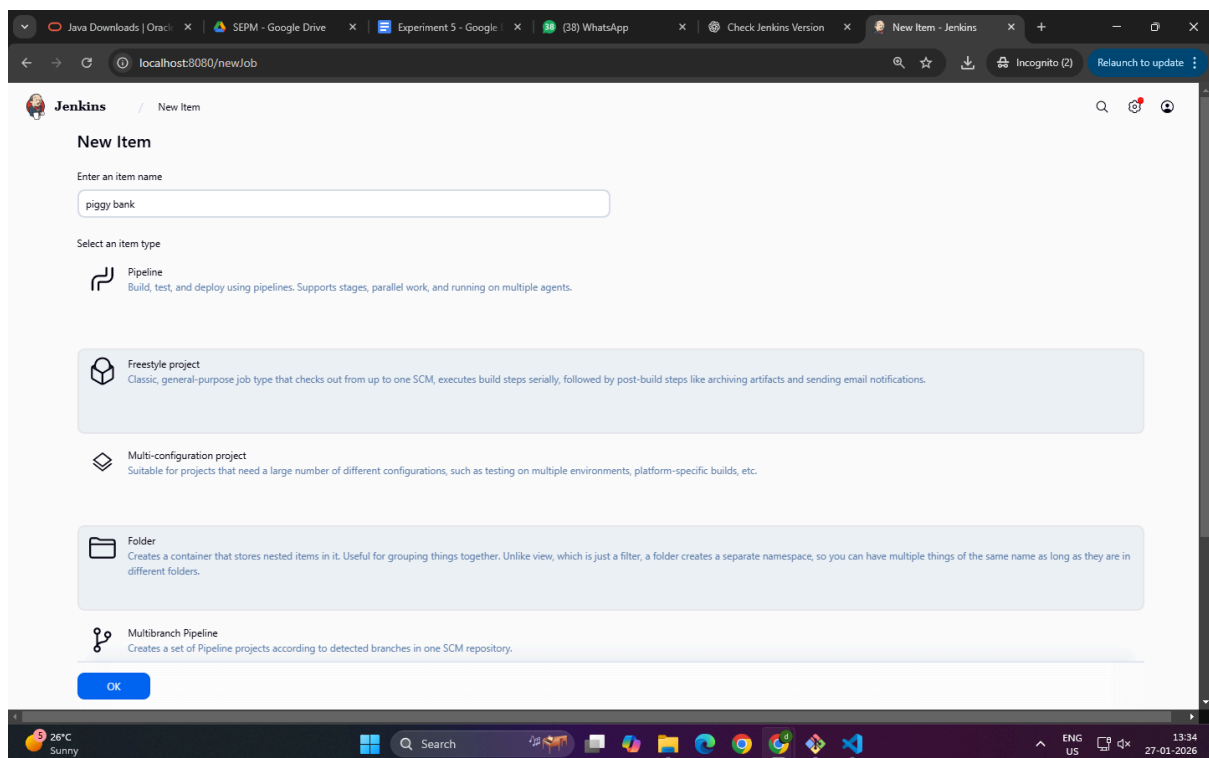
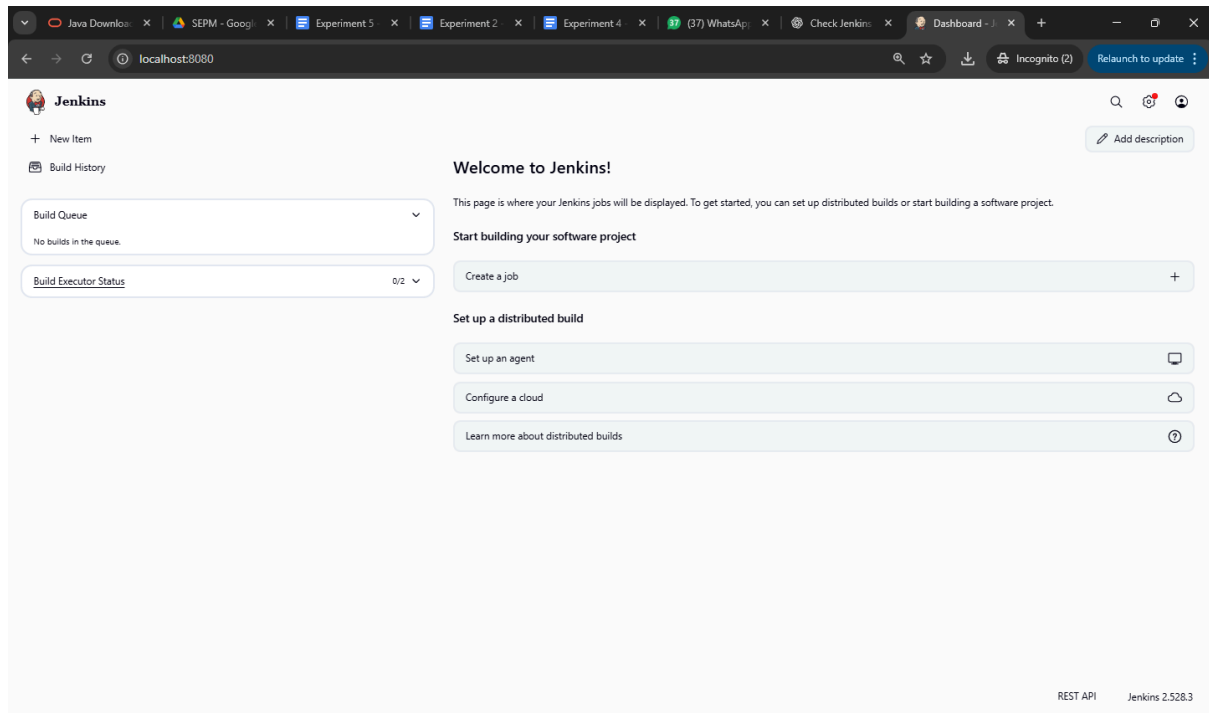
Parameterising the project with a string parameter as follows:

Configuring the build steps:

Setting the parameter for the build:

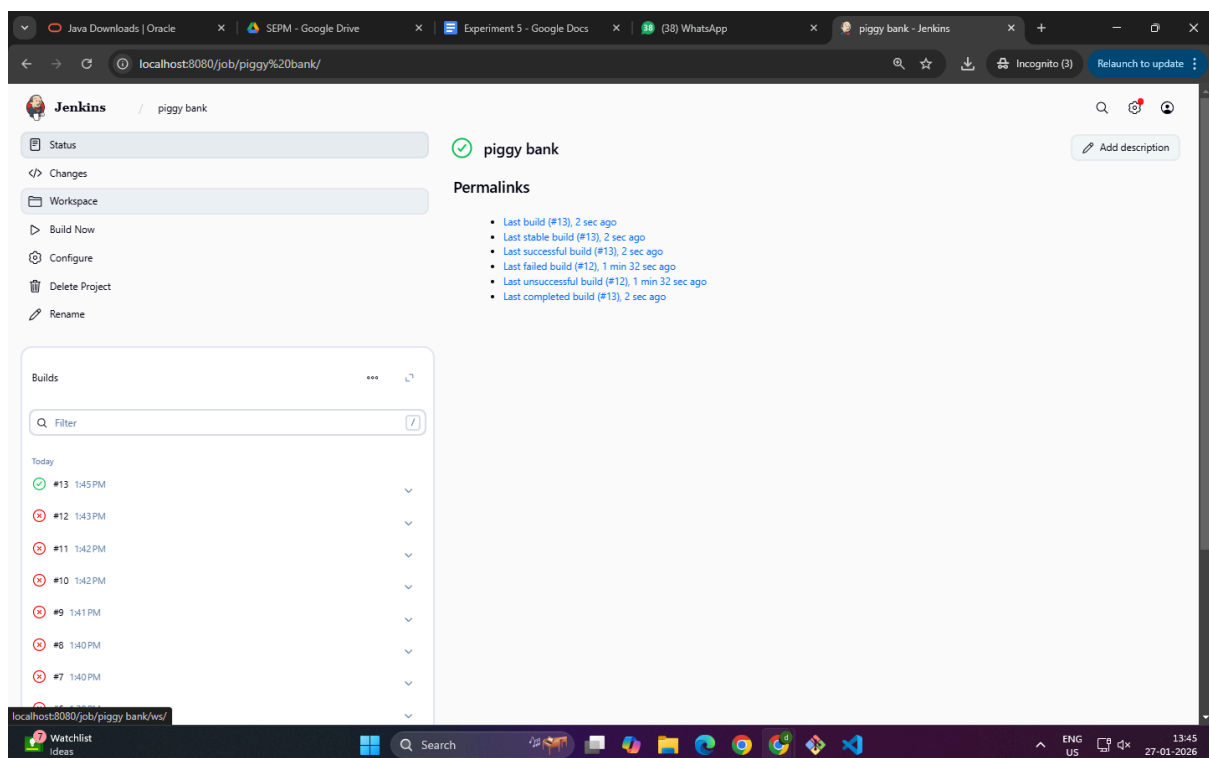
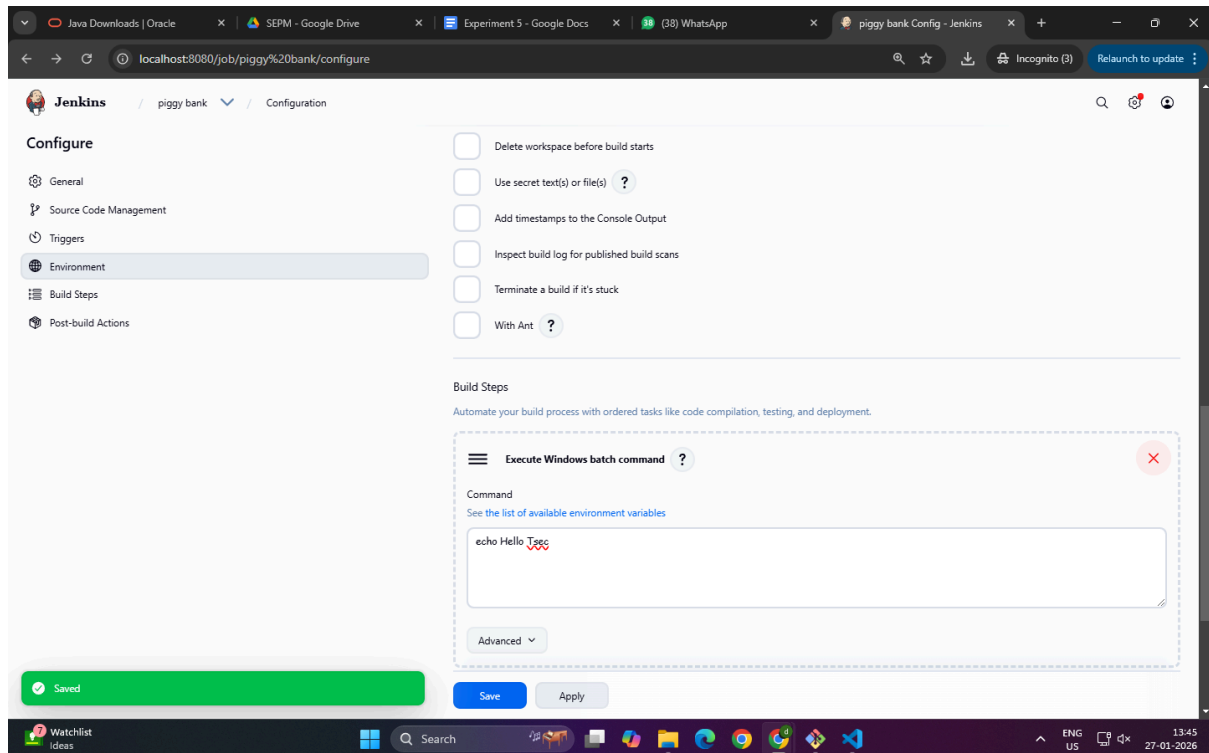
Console output after building:

Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.



The screenshot shows the Jenkins Configuration page for a job named 'piggy bank'. The 'General' tab is selected in the left sidebar. The main area is titled 'General' and has an 'Enabled' status with a blue checkmark. Under the 'Description' section, there is a large text area. Below it, there are several checkboxes: 'Discard old builds', 'GitHub project', 'This project is parameterized', 'Throttle builds', and 'Execute concurrent builds if necessary'. An 'Advanced' dropdown menu is visible. The 'Source Code Management' section is currently set to 'None'. At the bottom, there are 'Save' and 'Apply' buttons. The browser's address bar shows 'localhost:8080/job/piggy%20bank/configure'.

The screenshot shows the Jenkins Configuration page for the same job, but the 'Build Steps' tab is selected in the left sidebar. The main area is titled 'Build Steps' and contains a 'Command' section with a text area containing the command 'echo "Hello Tsec"'. Below this, there is an 'Advanced' dropdown menu and a '+ Add build step' button. The 'Post-build Actions' section is also visible, with a '+ Add post-build action' button. At the bottom, there are 'Save' and 'Apply' buttons. A green 'Saved' notification bar is visible at the bottom left. The browser's address bar shows 'localhost:8080/job/piggy%20bank/configure'.



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Batch: T21

Roll No.

The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL: `localhost:8080/job/piggy%20bank/lastBuild/console`. The Jenkins header shows the job name 'piggy bank' and build number '#13'. The left sidebar contains links for Status, Changes, Console Output (selected), Edit Build Information, Delete build #13, Timings, and Previous Build. The main area displays the 'Console Output' for build #13, which is successful. The output text is as follows:

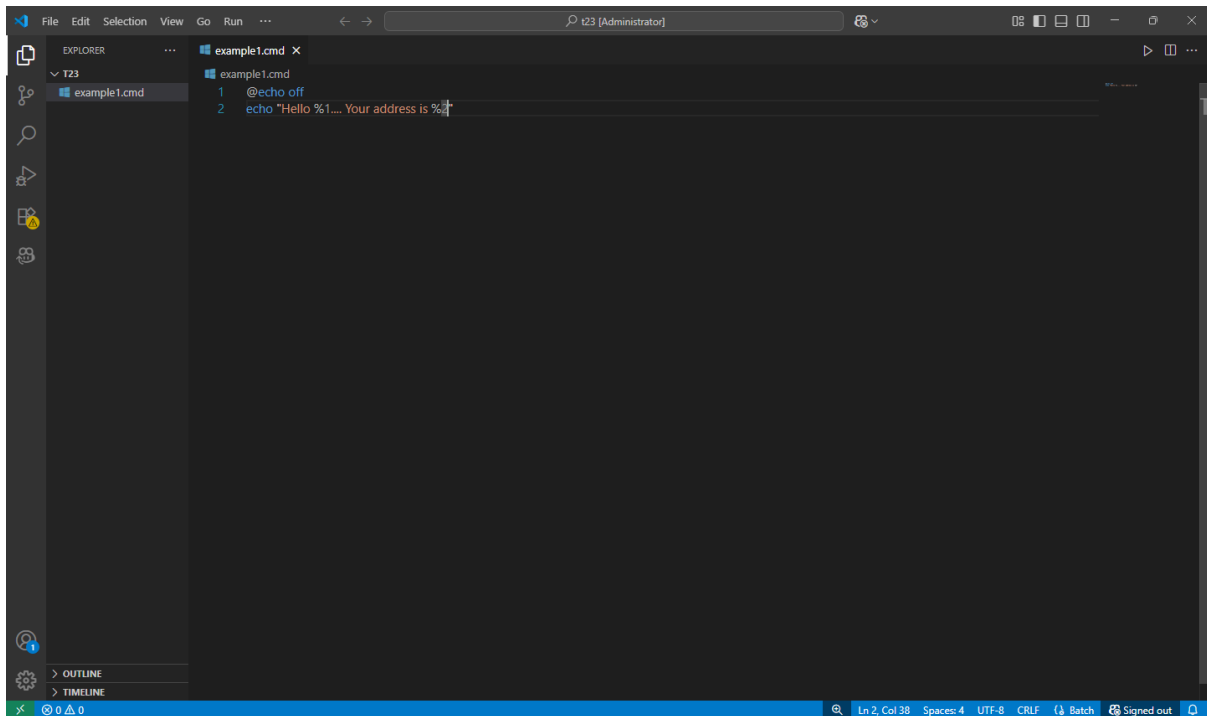
```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\workspace\piggy bank
[piggy bank] $ cmd /c call C:\WINDOWS\TEMP\Jenkins8909928918577008250.bat

C:\ProgramData\Jenkins\workspace\piggy bank>echo Hello Tsec
Hello Tsec

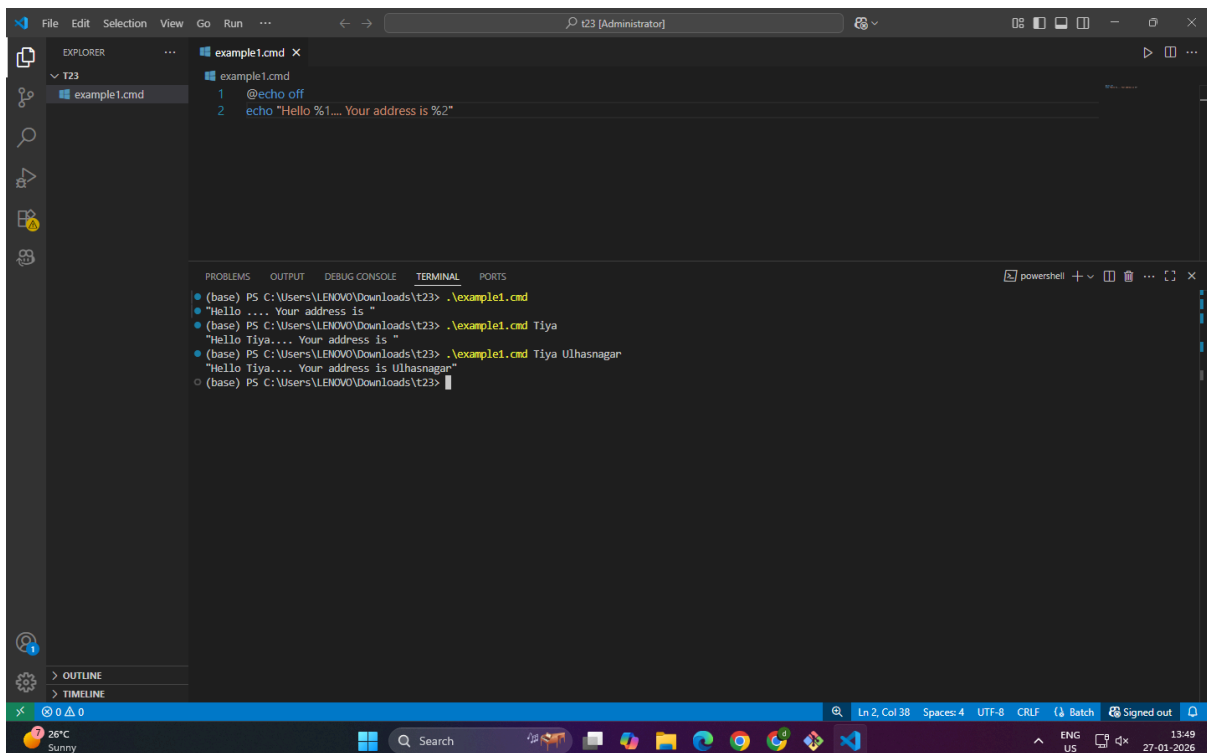
C:\ProgramData\Jenkins\workspace\piggy bank>exit 0
Finished: SUCCESS
```

At the bottom right of the Jenkins interface, it says 'REST API' and 'Jenkins 2.528.3'. The Windows taskbar is visible at the bottom of the screen.

1.2

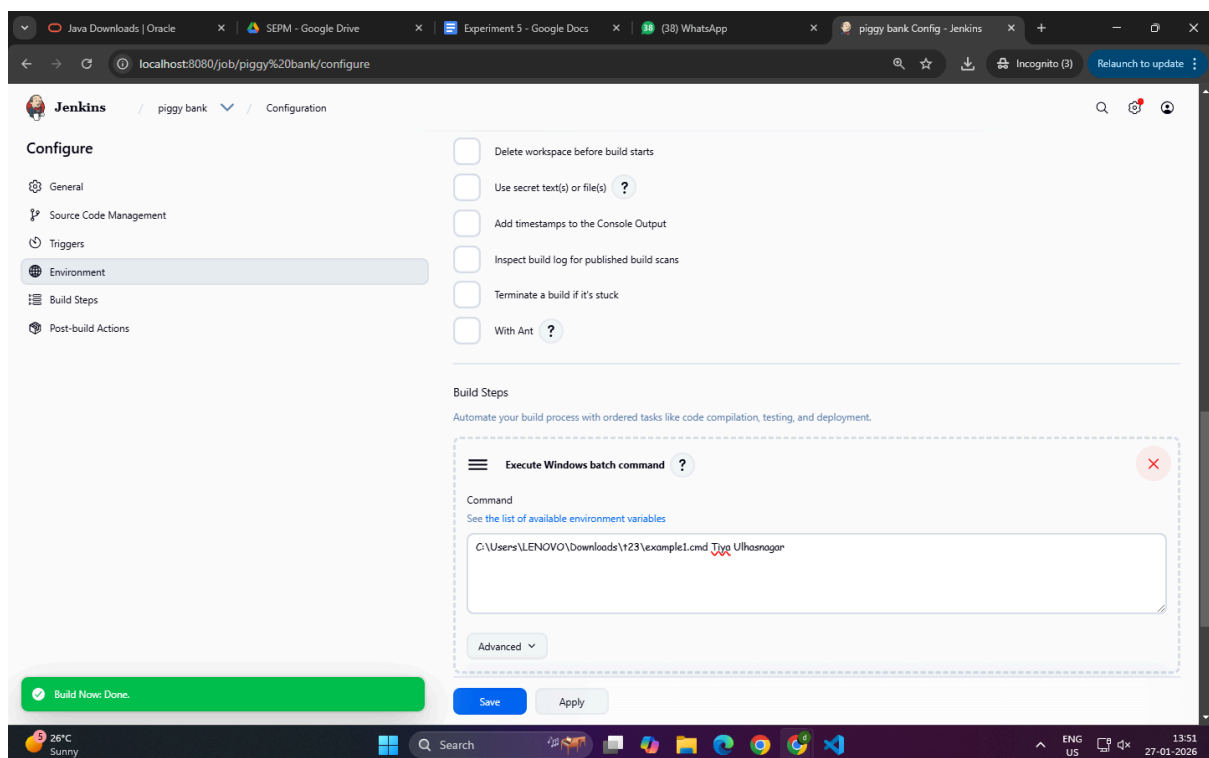
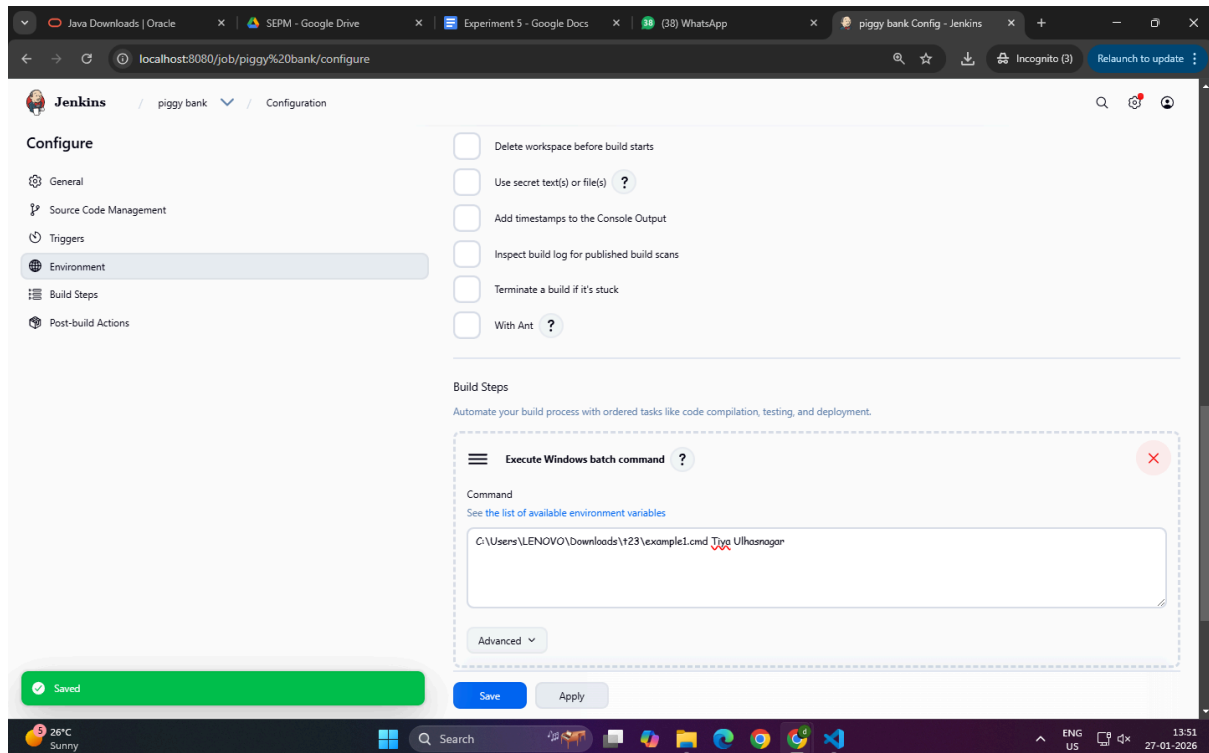


```
File Edit Selection View Go Run ... t23 [Administrator]
EXPLORER example1.cmd x
  T23
    example1.cmd
      1 @echo off
      2 echo "Hello %1.... Your address is %2"
```



```
File Edit Selection View Go Run ... t23 [Administrator]
EXPLORER example1.cmd x
  T23
    example1.cmd
      1 @echo off
      2 echo "Hello %1.... Your address is %2"

TERMINAL
  (base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd
  "Hello .... Your address is "
  (base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya
  "Hello Tiya.... Your address is "
  (base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya Ulhasnagar
  "Hello Tiya.... Your address is Ulhasnagar"
  (base) PS C:\Users\LENOVO\Downloads\t23>
```



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The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL `localhost:8080/job/piggy%20bank/lastBuild/console`. The Jenkins header shows the job name 'piggy bank' and build number '#15'. The left sidebar contains links for Status, Changes, Console Output (selected), Edit Build Information, Delete build #15, Timings, and Previous Build. The main area is titled 'Console Output' with a green checkmark icon and buttons for Download, Copy, and View as plain text. The console output text is as follows:

```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\piggy bank
[piggy bank] $ cmd /c call C:\WINDOWS\TEMP\jenkins15919224875403028413.bat

C:\ProgramData\Jenkins\jenkins\workspace\piggy bank>C:\Users\LENOVO\Downloads\t23\example1.cmd Tiya Ulhasnagar
"Hello Tiya.... Your address is Ulhasnagar"
Finished: SUCCESS
```

At the bottom right of the Jenkins interface, it says 'REST API' and 'Jenkins 2.528.3'. The Windows taskbar at the bottom shows the date and time as 13:51 on 27-01-2026, along with system icons for temperature (26°C Sunny) and network.

Example 2

The image displays two screenshots from a computer screen. The top screenshot shows the Jenkins web interface for configuring a job named 'Example 2'. The 'General' tab is selected, showing options like 'Discard old builds', 'GitHub project', and 'This project is parameterized'. The 'Source Code Management' section is set to 'None'. The bottom screenshot shows the Visual Studio Code editor with a Java file named 'Example2.java'. The code defines a class 'Example2' with a 'main' method that prints 'Hello World!'. The terminal at the bottom shows the command prompt in a PowerShell session.

Jenkins Configuration Screenshot:

- Browser: Incognito (3)
- URL: localhost:8080/job/Example%202/configure
- Page Title: Jenkins / Example 2 / Configuration
- Left Sidebar: Configure (General, Source Code Management, Triggers, Environment, Build Steps, Post-build Actions)
- General Tab: Enabled (checked)
- Description: Plain text (Preview)
- Options: ☐ Discard old builds, ☐ GitHub project, ☐ This project is parameterized, ☐ Throttle builds, ☐ Execute concurrent builds if necessary
- Source Code Management: ☒ None
- Buttons: Save, Apply

VS Code Editor Screenshot:

- File Explorer: T23 (example1.cmd, Example2.class, Example2.java, Main.class)
- Editor: Example2.java (Main > main(String[]))
- Code:

```
1 class Example2 {  
2     public static void main(String args[]) {  
3         System.out.println("Hello World!");  
4     }  
5 }
```
- Terminal: (base) PS C:\Users\LENOVO\Downloads\t23>

The image displays two screenshots of the Jenkins web interface, showing the configuration and overview of a job named 'Example 2'.

Top Screenshot: Jenkins Configuration Page

- URL:** `localhost:8080/job/Example%202/configure`
- Page Title:** Jenkins / Example 2 / Configuration
- Left Sidebar (Configure):**
 - General
 - Source Code Management
 - Triggers
 - Environment** (Selected)
 - Build Steps
 - Post-build Actions
- Main Content Area:**
 - Environment Section:** Includes checkboxes for 'Delete workspace before build starts', 'Use secret text(s) or file(s)', 'Add timestamps to the Console Output', 'Inspect build log for published build scans', 'Terminate a build if it's stuck', and 'With Ant'.
 - Build Steps Section:** Includes a description 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' and a step titled 'Execute Windows batch command'. The command field contains `C:\Users\LENOVO\Downloads\T23\Example2.java`.
- Buttons:** 'Save' and 'Apply' buttons are at the bottom.

Bottom Screenshot: Jenkins Overview Page

- URL:** `localhost:8080/job/Example%202/`
- Page Title:** Jenkins / Example 2
- Left Sidebar:**
 - Status** (Selected)
 - Changes
 - Workspace
 - Build Now
 - Configure
 - Delete Project
 - Rename
- Main Content Area:**
 - Example 2** (Job Name)
 - Permalinks** (Section Header)
 - Builds Section:** Shows a list of builds. The first build is labeled 'Today' and '1 2:00 PM'.
- Footer:** A green banner at the bottom indicates 'Build scheduled'.

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The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL `localhost:8080/job/Example%202/1/console`. The Jenkins header includes the logo and navigation links for 'Example 2' and '#1'. The left sidebar contains links for 'Status', 'Changes', 'Console Output' (selected), 'Edit Build Information', and 'Timings'. The main area is titled 'Console Output' and shows a progress bar. The output text is as follows:

```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example 2
[Example 2] $ cmd /c call C:\WINDOWS\TEMP\jenkins14948875696471621875.bat

C:\ProgramData\Jenkins\jenkins\workspace\Example 2>C:\Users\LENOVO\Downloads\t23\Example2.java
```

At the bottom of the console output, there are three dots indicating further output. The bottom of the browser window shows the REST API link and Jenkins version 2.528.3. The Windows taskbar at the very bottom shows the Nifty bank logo, search bar, and system clock indicating 14:01 on 27-01-2026.

Example 3

The image displays two screenshots of the Jenkins web interface, illustrating the setup of a new item.

Top Screenshot: New Item Page

- The browser address bar shows `localhost:8080/view/all/newJob`.
- The page title is "New Item".
- The "Enter an item name" field contains "Example 3".
- The "Select an item type" section shows several options:
 - Pipeline**: Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
 - Freestyle project**: Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications. (This option is highlighted).
 - Multi-configuration project**: Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
 - Folder**: Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
 - Multibranch Pipeline**: Creates a set of Pipeline projects according to detected branches in one SCM repository.
- An "OK" button is visible at the bottom.

Bottom Screenshot: Configuration Page

- The browser address bar shows `localhost:8080/job/Example%203/configure`.
- The page title is "Configure".
- The left sidebar shows the configuration tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The "General" tab is selected.
- The "This project is parameterized" checkbox is checked.
- A dropdown menu is open, showing the following parameter types:
 - Boolean Parameter
 - Choice Parameter
 - Credentials Parameter
 - File Parameter
 - Multi-line String Parameter
 - Password Parameter
 - Run Parameter
 - String Parameter
- The "Triggers" section is visible below the parameterization options, with the text "Set up automated actions that start your build based on specific events, like code changes or scheduled times."
- Buttons for "Save" and "Apply" are at the bottom.

The screenshot shows the Jenkins Configuration page for 'Example 3'. The left sidebar contains a 'Configure' section with a list of options: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected. The main content area shows a 'String Parameter' configuration. At the top, a blue checkmark indicates 'This project is parameterized'. The 'String Parameter' section has a 'Name' field with the value 'fname', an empty 'Default Value' field, and an empty 'Description' field. Below these fields are links for 'Plain text' and 'Preview', and a checkbox for 'Trim the string'. At the bottom of the configuration area are 'Save' and 'Apply' buttons. The browser's address bar shows 'localhost:8080/job/Example%203/configure'. The Windows taskbar at the bottom shows the system clock as 14:04 on 27-01-2026.

The screenshot shows the Jenkins Configuration page for 'Example 3', specifically the 'Choice Parameter' configuration. The left sidebar is the same as the previous screenshot. The main content area shows the 'Choice Parameter' section. It has a 'Name' field with the value 'City', a 'Choices' field with a list of options: 'Bandra', 'Juhu', 'Kalyan', and 'Mulund', and an empty 'Description' field. Below these fields are links for 'Plain text' and 'Preview', and a checkbox for 'Trim the string'. At the bottom of the configuration area are 'Save' and 'Apply' buttons. The browser's address bar shows 'localhost:8080/job/Example%203/configure'. The Windows taskbar at the bottom shows the system clock as 14:04 on 27-01-2026.

The image shows two screenshots from a computer screen. The top screenshot is of the Jenkins web interface, specifically the 'Configure' page for a job named 'Example 3'. The 'General' tab is selected in the left sidebar. A green 'Saved' message is visible at the bottom left. The main area shows a 'String Parameter' configuration with the name 'fname' and a 'Choice Parameter' configuration below it. The bottom screenshot is of the Visual Studio Code editor. The Explorer sidebar on the left shows a project named 'T23' with files 'example1.cmd', 'Example2.class', 'Example2.java', 'example3.cmd', and 'Main.class'. The main editor window shows the content of 'example3.cmd', which contains two lines: '@echo off' and 'echo "Hello %1.... Your city is %2"'. The bottom status bar of VS Code shows 'Ln 2, Col 29', 'Spaces: 4', 'UTF-8', 'CRLF', and 'Batch' mode.

Jenkins Configuration Screenshot:

- Page: Jenkins / Example 3 / Configuration
- Tab: General
- Buttons: Source Code Management, Triggers, Environment, Build Steps, Post-build Actions
- Message: This project is parameterized
- Parameter Type: String Parameter
- Name: fname
- Default Value: (empty)
- Description: (empty)
- Trim the string: (unchecked)
- Parameter Type: Choice Parameter
- Name: (empty)
- Status: Saved

VS Code Editor Screenshot:

- File: example3.cmd
- Content:

```
1 @echo off
2 echo "Hello %1.... Your city is %2"
```
- Terminal: (base) PS C:\Users\LENOVO\Downloads\t23>
- Status Bar: Ln 2, Col 29, Spaces: 4, UTF-8, CRLF, Batch, Signed out

The image displays two screenshots of the Jenkins web interface, showing the configuration and build process for a project named 'Example 3'.

Top Screenshot: Jenkins Configuration Page

- URL:** `localhost:8080/job/Example%203/configure`
- Page Title:** Jenkins / Example 3 / Configuration
- Left Sidebar (Configure):**
 - General
 - Source Code Management
 - Triggers
 - Environment
 - Build Steps**
 - Post-build Actions
- Build Steps:**
 - Automate your build process with ordered tasks like code compilation, testing, and deployment.
 - Step: **Execute Windows batch command** (with a red 'X' icon)
 - Command: `C:\Users\LENOVO\Downloads\t23\example3.cmd`
 - Advanced options: **Advanced** (dropdown)
 - + Add build step
- Post-build Actions:**
 - Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.
 - + Add post-build action
- Buttons:** **Save** (blue), **Apply** (grey)
- Status:** **Saved** (green bar)
- REST API:** Jenkins 2.528.3

Bottom Screenshot: Jenkins Build Page

- URL:** `localhost:8080/job/Example%203/build?delay=0sec`
- Page Title:** Jenkins / Example 3
- Left Sidebar (Build):**
 - Status
 - Changes
 - Workspace
 - Build with Parameters
 - Configure**
 - Delete Project
 - Rename
- Project Example 3:**
 - This build requires parameters:
 - fname:
 - City:
 - Build** (green button), **Cancel** (grey button)
- Builds:**
 - No builds
- Status:** Jenkins 2.528.3

Windows Taskbar (Common to both): 26°C Sunny, Search, Taskbar icons, System tray: ENG US, 14:07, 27-01-2026.

The image displays two screenshots of the Jenkins web interface, showing the configuration and execution of a job named 'Example 3'.

Top Screenshot: Jenkins Job Overview

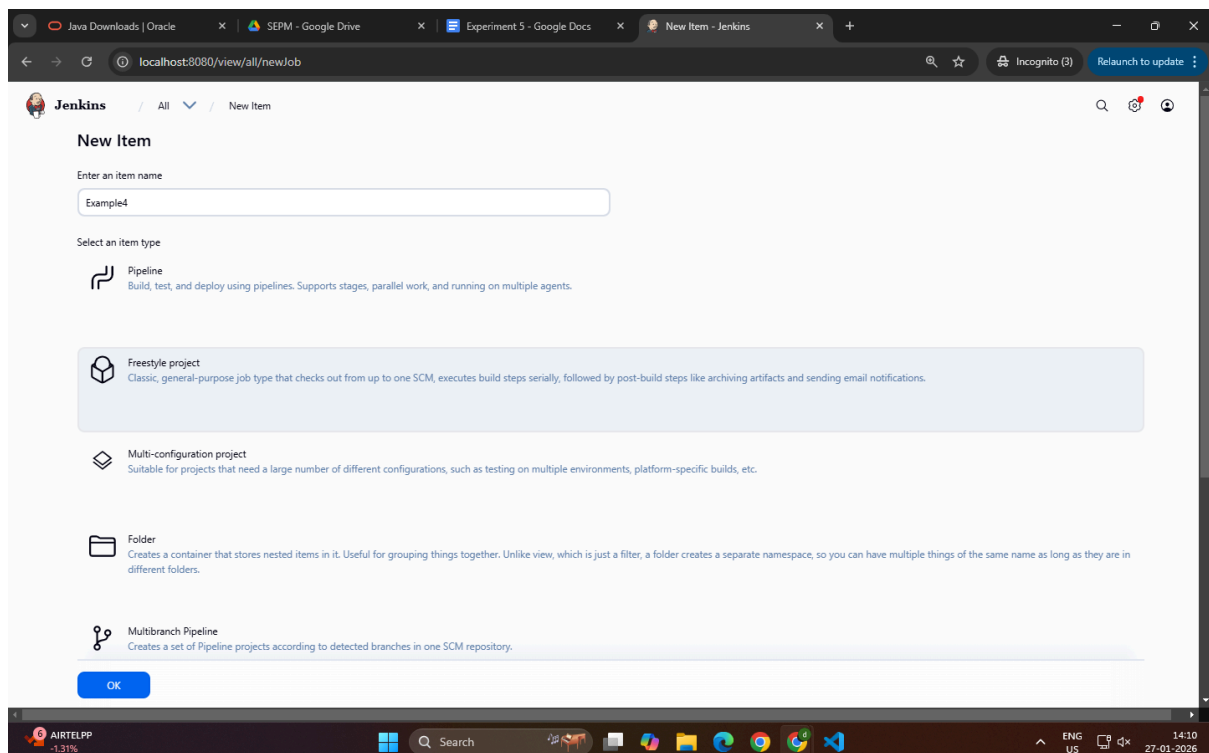
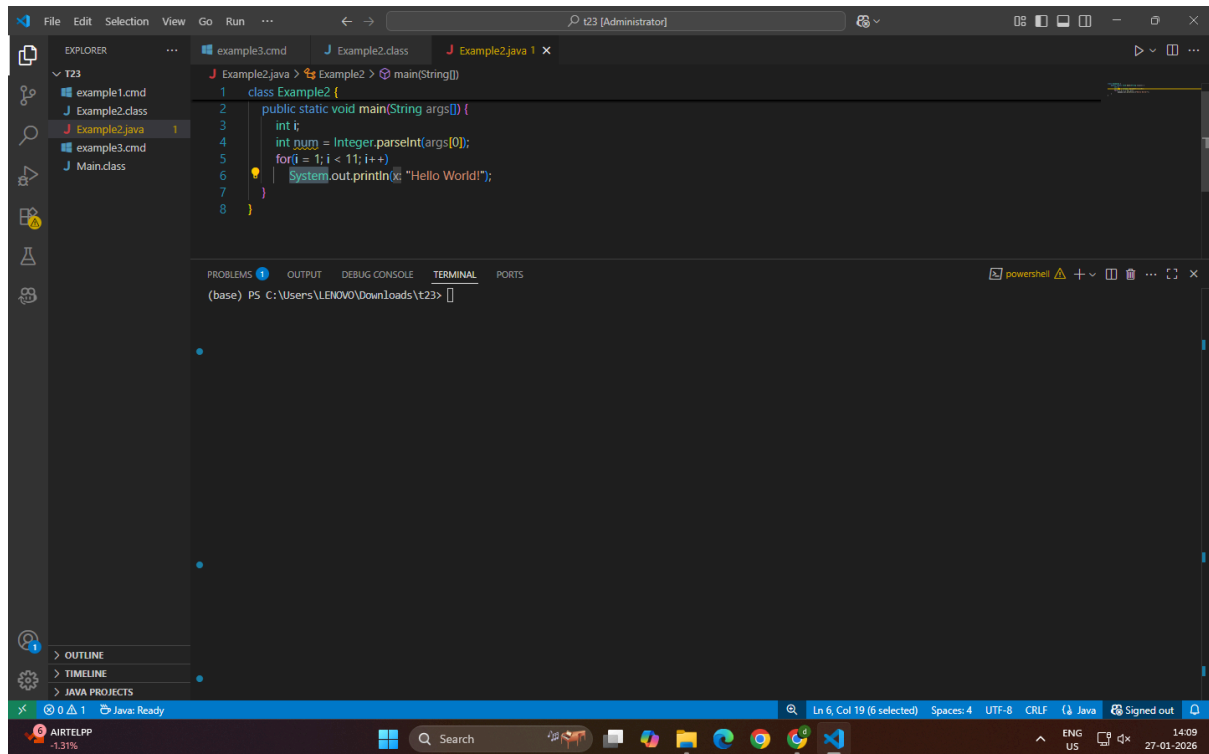
- URL:** `localhost:8080/job/Example%203/`
- Job Name:** Example 3
- Left Sidebar:** Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, Rename.
- Permalinks:** Last build (#1), 8 ms ago
- Builds:** A table showing build #1 at 2:08 PM, with a status icon indicating success.
- REST API:** Jenkins 2.528.3

Bottom Screenshot: Console Output

- URL:** `localhost:8080/job/Example%203/1/console`
- Job Name:** Example 3
- Build:** #1
- Section:** Console Output
- Actions:** Download, Copy, View as plain text
- Output:**

```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example 3
[Example 3] $ cmd /c call C:\WINDOWS\TEMP\jenkins5651314562182551435.bat
C:\ProgramData\Jenkins\jenkins\workspace\Example 3>C:\Users\LENOVO\Downloads\t23\example3.cmd
"Hello .... Your city is "
Finished: SUCCESS
```
- REST API:** Jenkins 2.528.3

3.2



The screenshot shows the Jenkins Configuration page for a job named 'Example4'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected. In the main area, there is a checkbox for 'GitHub project' which is unchecked. Below it, a checkbox 'This project is parameterized' is checked. A 'String Parameter' configuration box is visible, with the 'Name' field set to 'num'. The 'Default Value' and 'Description' fields are empty. There is a 'Trim the string' checkbox which is unchecked. At the bottom of the configuration box is a '+ Add Parameter' button. Below the configuration box are 'Save' and 'Apply' buttons. The browser's address bar shows 'localhost:8080/job/Example4/configure'. The Windows taskbar at the bottom shows the system clock as 14:11 on 27-01-2026.

The screenshot shows the Jenkins Configuration page for a job named 'Example4', specifically the 'Build Steps' tab. The left sidebar is the same as in the previous screenshot, but the 'Build Steps' tab is now selected. The main area has a heading 'Build Steps' with a subtext 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' Below this, there is an 'Execute Windows batch command' configuration box. The 'Command' field contains the text 'C:\Users\LENOVO\Downloads\123\Example4.java %num%'. There is an 'Advanced' dropdown menu below the command field. At the bottom of the configuration box is a '+ Add build step' button. Below the configuration box, there is a 'Post-build Actions' section with a subtext 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' and a '+ Add post-build action' button. At the bottom of the page are 'Save' and 'Apply' buttons. A green 'Saved' notification bar is visible at the bottom left. The browser's address bar shows 'localhost:8080/job/Example4/configure'. The Windows taskbar at the bottom shows the system clock as 14:12 on 27-01-2026.

The image displays two screenshots of the Jenkins web interface, showing the configuration and execution of a build job.

Top Screenshot: Project Configuration

- URL:** `localhost:8080/job/Example4/build?delay=0sec`
- Page Title:** Project Example4
- Left Sidebar:** Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, Rename.
- Main Content:** This build requires parameters. A parameter named `num` is shown with a value of `4`. Buttons for **Build** and **Cancel** are visible.
- Builds List:** A table showing build history. The first build is labeled `#1` and occurred at `2:12 PM`.
- Footer:** Jenkins 2.528.3

Bottom Screenshot: Console Output

- URL:** `localhost:8080/job/Example4/lastBuild/console`
- Page Title:** Console Output
- Left Sidebar:** Status, Changes, Console Output, Edit Build Information, Parameters, Timings.
- Main Content:** The console output shows the build process. It starts with `Started by user Tiya Nathwani`, followed by `Running as SYSTEM`. The build is executed in the workspace `C:\ProgramData\Jenkins\jenkins\workspace\Example4`. The command `[Example4] $ cmd /c call C:\WINDOWS\TEMP\jenkins5789436201230215465.bat` is shown. The output of the command is `C:\ProgramData\Jenkins\jenkins\workspace\Example4>C:\Users\LENOVO\Downloads\t23\Example4.java`.
- Footer:** REST API, Jenkins 2.528.3